Topic 1: Measuring the Economy

GDP, Inflation, and Unemployment

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Introduction to Economics II
Based on Mankiw & Taylor, Chapters 20-22

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Roadmap

What will we cover in this topic?

Part A: Introduction

- Why study macroeconomics?
- Long run vs short run: a crucial distinction

Part B: The three main macroeconomic indicators

1. **GDP** – Measuring total production
2. **Inflation** – Measuring rising prices
3. **Unemployment** – Measuring unused labor

These are the **foundation** for everything else in macroeconomics!

Why Study Macroeconomics?

Why do we need a different approach to study the economy?

In Economics I (Microeconomics):

- Individual decisions: How much coffee should I buy?
- Specific markets: What determines the price of apartments?
- Tools: Supply and demand, consumer choice, firm behavior

But some questions are different:

- Why are there 3 million unemployed in Spain?
- Why did all prices rise 8% in 2022?
- Why is Germany richer than Greece?

These require studying the economy **as a whole**

The Big Macroeconomic Questions

Questions we will learn to analyze:

1. **Growth:** Why does production grow over time?
   Why faster in China than in Sub-Saharan Africa?
2. **Fluctuations:** Why do recessions happen?
   The 2008 financial crisis, the 2020 COVID recession
3. **Unemployment:** Why can't everyone find a job?
   Spain $\sim$12% vs Germany $\sim$3%
4. **Inflation:** Why do prices rise?
   What happened in 2022? What can central banks do?
5. **Policy:** Can governments improve economic outcomes?
   Fiscal policy, monetary policy, trade policy

Mankiw's Three Macroeconomic Principles

From the *Ten Principles of Economics*:

8. **A country's standard of living depends on its ability to produce goods and services**

   → Topics 2-3: Production and Growth

9. **Prices rise when the government prints too much money**

   → Topic 5: Money and Inflation

10. **Society faces a short-run trade-off between inflation and unemployment**

    → Topics 7-8: Business Cycles and Stabilization Policy

This course will explain these three principles

The Long Run vs The Short Run

A crucial distinction in macroeconomics

Long Run

- Prices are **flexible**
- Markets clear (supply = demand)
- Economy at full employment
- Focus: economic growth, living standards

Classical approach (Adam Smith, 1776)

Short Run

- Some prices are **sticky**
- Markets may not clear
- Unemployment can persist
- Focus: business cycles, recessions

Keynesian approach (J.M. Keynes, 1936)

This is NOT the same as micro's long run / short run!

In micro: time for firms to adjust capacity. In macro: time for **prices** to adjust.

Why Does This Distinction Matter?

The key question since 1776: Does a market economy naturally tend toward full employment?

Classical view (long run):

- If there's unemployment, wages will fall
- Lower wages $\rightarrow$ firms hire more workers
- Eventually, everyone who wants a job finds one
- **Conclusion:** Markets self-correct. No need for government intervention.

Keynesian view (short run):

- Wages and prices don't adjust instantly
- The Great Depression (1929): 25% unemployment for years
- Markets can get "stuck" in bad equilibria
- **Conclusion:** Government policy may be needed to restore full employment.

[$\rightarrow$ *Topic 7-8*] We'll explore this debate in detail later in the course

Flexible vs Sticky Prices: An Illustration

When prices are flexible, markets clear automatically:

- Excess supply? → Price falls → Quantity demanded increases → Balance restored
- Excess demand? → Price rises → Quantity supplied increases → Balance restored

When prices are sticky, quantities must adjust instead:

- Wages too high? → Firms can't afford all workers → **Unemployment**
- Prices too high? → Consumers can't buy all goods → **Unsold inventory**
- Firms respond by cutting production → **Recession**

Key insight: When prices don't adjust, **quantities must adjust instead**.

This is why recessions cause unemployment, not just lower wages!

What Economists Agree and Disagree On

Economists mostly agree on the long run:

- Countries that invest more in education grow faster
- Inflation is ultimately a monetary phenomenon
- Trade benefits countries in the long run

Economists disagree on the short run:

- Should governments run deficits during recessions?
- Should central banks raise or lower interest rates now?
- Do minimum wages increase unemployment?

Why the disagreement?

- Different views on how quickly prices adjust
- Different views on how expectations are formed
- Different values about government's role

What is GDP?

How do we measure the size of an economy?

Gross Domestic Product (GDP): The market value of all final goods and services produced within a country during a given period.

Let's break it down:

Market value Uses prices as weights (need common unit)

Final goods Sold to end users (not intermediate inputs)

Produced New production only (not used goods or transfers)

Within a country Geographic criterion (not nationality)

Given period Flow concept (usually one year or quarter)

GDP vs GNP

Two ways to measure national output:

GDP (Gross Domestic Product) Production within a country's borders

Geographic criterion: Where is it produced?

GNP (Gross National Product) Production by a country's residents

Nationality criterion: Who produces it?

Relationship:

$$\text{GNP} = \text{GDP} - \text{Income of foreigners in Spain} + \text{Income of Spaniards abroad}$$

In practice: GDP and GNP are usually very close.
Exception: Countries with many workers abroad (e.g., Philippines) or many foreign companies (e.g., Ireland)

We will focus on **GDP** throughout this course.

The Expenditure Approach

How do we calculate GDP?

GDP is measured by adding up all **expenditures** on final goods:

$$\boxed{Y = C + I + G + NX}$$

C - Consumption Household spending on goods and services

I - Investment Business fixed investment + residential investment + inventory changes

G - Government Government purchases of goods and services

NX - Net Exports Exports minus imports $(X - M)$

This is the **fundamental macroeconomic identity**

Components of GDP: Details

Consumption (C) – About 60% of GDP

- Durable goods (cars, appliances)
- Non-durable goods (food, clothing)
- Services (healthcare, education, restaurants)

Investment (I) – About 20% of GDP

- Business fixed investment (equipment, structures)
- Residential investment (new houses)
- Inventory changes (stocks of goods)

Government purchases (G) – About 20% of GDP

- Goods and services bought by government
- **NOT transfers** (pensions, unemployment benefits)

Net exports (NX) – Close to 0% in Spain

- Exports (X) minus Imports (M)

What DOES Count in GDP?

These count: ✓

- New car sold by dealer
- Haircut at the barber
- New house construction
- Government pays teacher's salary
- Spanish wine exported to France

Common pattern:

- Market transaction
- For a final good or service
- Produced this period
- Within the country's borders

What Does NOT Count in GDP?

These do NOT count: $\times$

- **Used goods** (second-hand car) – already counted when new
- **Intermediate goods** (flour used by bakery) – would double-count
- **Financial transactions** (buying stocks) – not production
- **Transfer payments** (pensions, unemployment benefits) – no production in exchange
- **Household production** (cooking at home, childcare) – not market transaction
- **Underground economy** (illegal activities, unreported work) – hidden from statistics

Key principle: GDP measures current *production* of goods and services sold in markets

Example: Production Chain

Question: A farmer sells wheat to a baker for €100. The baker makes bread and sells it for €180. What is GDP?

Wrong answer: €100 + €180 = €280 **(This double-counts!)**

Correct answer: €180 (Only the final good counts)

Alternative calculation – Value Added:

- Farmer's value added: €100 - €0 = €100
- Baker's value added: €180 - €100 = €80
- Total GDP = €100 + €80 = €180 ✓

Both methods give the same answer!

Real vs Nominal GDP

Are we producing more, or are prices just rising?

Nominal GDP

- Values output at **current prices**
- Changes due to:
  - More production, OR
  - Higher prices
- Not a good measure of economic growth

Real GDP

- Values output at **constant prices** (base year)
- Changes only due to:
  - More production
- Measures actual economic growth
- Used to compare over time

Real GDP removes the effect of inflation

Simple Example: Pizza Economy

Imagine an economy that produces only pizza:

Year	Quantity	Price	Nominal GDP
2023	100 pizzas	€10	€1,000
2024	110 pizzas	€12	€1,320

Nominal GDP growth: $(1320 - 1000)/1000 = 32\%$

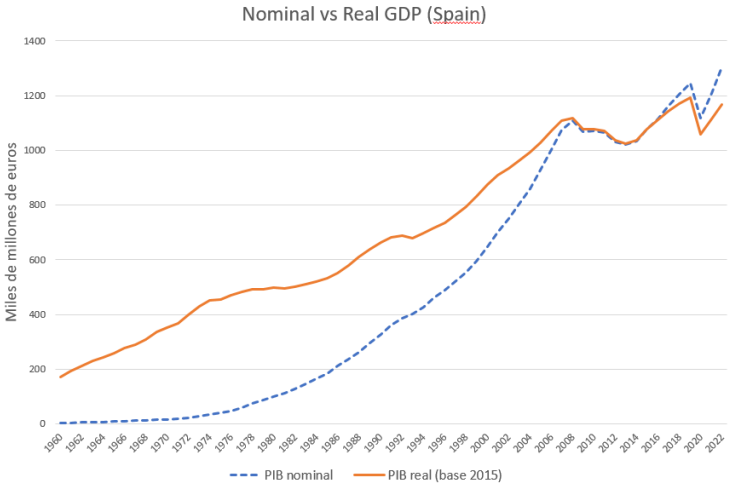
But this overstates real growth! Prices went up too.

Real GDP (using 2023 as base year):

- 2023: $100 \times €10 = €1{,}000$
- 2024: $110 \times €10 = €1{,}100$ (using 2023 prices)

Real GDP growth: $(1100 - 1000)/1000 = 10\%$ $\leftarrow$ True growth!

Nominal vs Real GDP: Spain



The GDP Deflator

How do we measure the overall price level?

The **GDP Deflator** measures the price level of all goods produced:

$$\boxed{\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100}$$

Interpretation:

- Base year: GDP Deflator $= 100$
- If GDP Deflator $= 110 \rightarrow$ Prices are 10% higher than base year
- If GDP Deflator $= 95 \rightarrow$ Prices are 5% lower than base year

Inflation rate: % change in GDP Deflator (or CPI)

Calculating Real GDP Growth

Spain's GDP (example):

Year	Nominal GDP (billions €)	GDP Deflator (2015=100)	Real GDP (billions 2015€)
2023	1,400	115.0	1,217
2024	1,470	118.0	1,246

Nominal growth: $(1470 - 1400)/1400 = 5.0\%$
Real growth: $(1246 - 1217)/1217 = 2.4\%$
Inflation: $(118.0 - 115.0)/115.0 = 2.6\%$

Notice: Nominal growth $\approx$ Real growth + Inflation

GDP and Economic Well-Being

What GDP measures well: ✓

- Market production of goods and services
- Material standard of living
- Economic activity

What GDP does NOT capture: ×

- **Household production** (cooking, childcare, home repairs)
- **Leisure time** (working less may reduce GDP but increase welfare)
- **Environmental quality** (pollution, resource depletion)
- **Income distribution** (GDP per capita hides inequality)
- **Underground economy** (unreported work, illegal activities)
- **Quality of goods** (improvements not fully captured)

GDP is useful but incomplete as a measure of well-being

Summary: GDP

Key concepts:

1. **GDP** $= C + I + G + NX$ (expenditure approach)
2. Only **final goods**, **newly produced**, count
3. **Real GDP** removes inflation, measures actual growth
4. **GDP Deflator** = (Nominal GDP / Real GDP) $\times$ 100
5. GDP is useful but doesn't capture everything about well-being

[$\rightarrow$ *Topic 2-8*] We'll use GDP as the main measure of output (Y) in all our models

Next: How do we measure the **price level** more directly?

Measuring the Cost of Living

How much more expensive is life today?

Two main price indices:

GDP Deflator Prices of all goods **produced** domestically

CPI (Consumer Price Index) Prices of goods **consumers buy**

Why CPI matters:

- Measures cost of living for households
- Used to adjust wages, pensions, contracts
- Most common inflation measure in the news

CPI: The Four Steps

How statistical agencies construct the CPI:

1. **Fix the basket:** Survey consumers to determine typical purchases
2. **Collect prices:** Find prices of all basket items (monthly)
3. **Calculate basket cost:** Multiply quantities $\times$ prices, sum up
4. **Compute the index:** Compare to base year, multiply by 100

Let's see an example...

CPI Example: Steps 1 & 2

Step 1 – Fix the basket:
A typical consumer buys: **4 loaves of bread** and **2 kg of bananas**

Step 2 – Collect prices each year:

Year	**Price of bread** (€/loaf)	**Price of bananas** (€/kg)
2022	1.00	1.50
2023	1.20	1.60
2024	1.40	1.80

Note: The basket (4 bread + 2 kg bananas) stays **fixed** across years

CPI Example: Step 3

Step 3 – Calculate the cost of the basket each year:

Cost = (4 loaves × price of bread) + (2 kg × price of bananas)

Year	Bread cost	Banana cost	Total basket
2022	$4 \times 1.00 = 4.00$	$2 \times 1.50 = 3.00$	**7.00 €**
2023	$4 \times 1.20 = 4.80$	$2 \times 1.60 = 3.20$	**8.00 €**
2024	$4 \times 1.40 = 5.60$	$2 \times 1.80 = 3.60$	**9.20 €**

The same basket costs more each year → prices are rising

CPI Example: Step 4

Step 4 – Compute the index (using 2022 as base year):

$$\text{CPI} = \frac{\text{Cost of basket in current year}}{\text{Cost of basket in base year}} \times 100$$

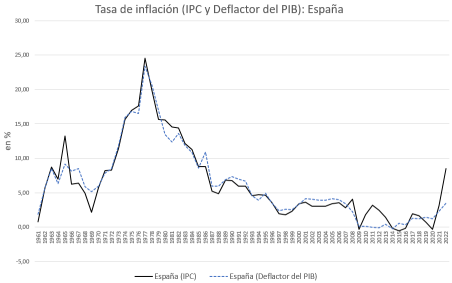
Year	Basket cost	CPI	Inflation rate
2022 (base)	7.00 €	$(7.00/7.00) \times 100 = 100.0$	—
2023	8.00 €	$(8.00/7.00) \times 100 = 114.3$	14.3%
2024	9.20 €	$(9.20/7.00) \times 100 = 131.4$	15.0%

Inflation rate = % change in CPI from previous year

Example 2024: $(131.4 - 114.3)/114.3 \times 100 = 15.0\%$

CPI vs GDP Deflator

	CPI	**GDP Deflator**
Goods included	Consumer basket	All domestic production
Imports?	Yes	No
Capital goods?	No	Yes
Basket	Fixed	Variable



Problems with CPI

CPI tends to overstate true inflation:

1. **Substitution bias:** Consumers switch to cheaper alternatives when prices rise, but the fixed basket doesn't capture this
2. **New products:** New goods (smartphones, streaming) improve life but aren't in old baskets
3. **Quality improvements:** A 2024 car is better than a 2004 car at similar price

Estimated bias: $\sim$1% per year (Lebow & Rudd, 2003)

Statistical agencies update baskets periodically to reduce bias

Using CPI: Adjusting Values Over Time

Converting euros from one year to another:

$$\boxed{\text{Value in Year B} = \text{Value in Year A} \times \frac{\text{CPI}_B}{\text{CPI}_A}}$$

Example: A worker earned €20,000 in 2015. What's that worth in 2024?

- CPI 2015 = 100 (base year)
- CPI 2024 = 125

€20,000 $\times \frac{125}{100} =$ **€25,000**

The worker needs €25,000 in 2024 to have the same **purchasing power**

Real vs Nominal Interest Rates

What is the true return on savings?

Nominal interest rate (i) What the bank pays you
Real interest rate (r) Your gain in purchasing power

Fisher Equation:

$$\boxed{r = i - \pi}$$

where π is the inflation rate

Example:

- Nominal rate: $i = 5\%$
- Inflation: $\pi = 3\%$
- Real rate: $r = 5\% - 3\% = 2\% \leftarrow$ What matters for decisions

[$\rightarrow$ *Topic 5, 8*] Real interest rate crucial for investment and saving decisions

Summary: Inflation

Key concepts:

1. **CPI** measures cost of a fixed basket of consumer goods
2. **Inflation** = % change in CPI (or GDP Deflator)
3. CPI vs GDP Deflator: Different baskets and coverage
4. CPI has biases → overstates inflation slightly
5. Use CPI to adjust values across time
6. **Real interest rate** = Nominal rate - Inflation

Next: How do we measure **unemployment**?

Who counts as unemployed?

Basic definitions:

Labor force	Employed + Unemployed
Unemployed	Without job, available, actively seeking
Not in labor force	Not working, not looking

Key rates:

$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labor force}} \times 100$$

$$\text{Participation rate} = \frac{\text{Labor force}}{\text{Working-age pop.}} \times 100$$

Who Counts as Unemployed?

To be **unemployed**, must satisfy **ALL three**:

1. Not working
2. Available to work
3. Actively seeking work

Examples:

- Laid off, actively applying → Unemployed
- Student not looking → Not in labor force
- Discouraged worker (stopped looking) → Not in labor force
- Part-time wanting full-time → Employed (underemployed)

Official rate may not capture discouraged workers or underemployment

Example: Spain (2024)

Numbers:

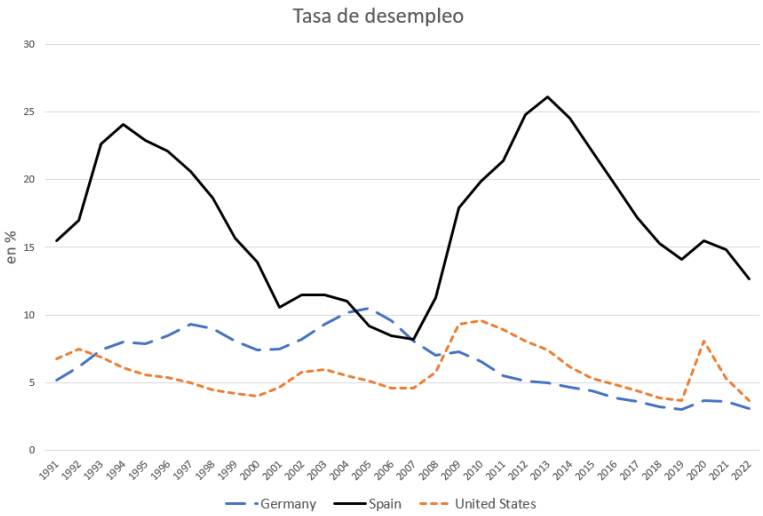
- Working-age: 40M
- Labor force: 24M
- Employed: 21M
- Unemployed: 3M

Rates:

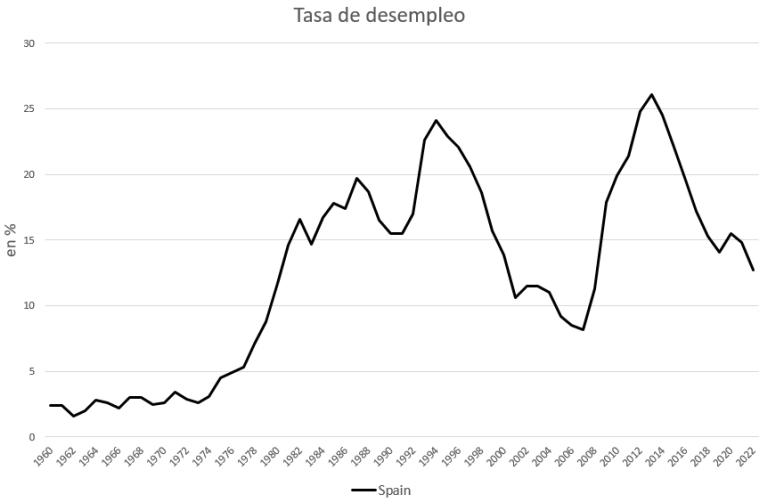
- Participation: 60%
- Unemployment: 12.5%
- Employment: 52.5%

Spain has historically higher unemployment than most EU countries

Unemployment: International Comparison



Unemployment: Spain Historical



Natural Rate of Unemployment

Natural rate ($\bar{u}$): Long-run unemployment rate when economy is at full employment

Key features:

- Also called NAIRU (Non-Accelerating Inflation Rate of Unemployment)
- Varies by country: US ∼4-5%, Spain ∼10-12%
- Not zero even in healthy economies
- [→ *Topic 8*] Central to Phillips curve analysis

Components: Natural rate = Frictional + Structural

Why not zero? Job matching takes time, skills don't always match jobs available

Types of Unemployment

1. **Frictional unemployment**
   - Workers between jobs, new graduates searching
   - Normal and short-term
   - Example: Recent graduate looking for first job

2. **Structural unemployment**
   - Long-term mismatch between skills and jobs
   - Due to technological change, industry shifts, location
   - Example: Coal miners when mines close

3. **Cyclical unemployment**
   - Due to economic downturns (recessions)
   - Disappears when economy recovers
   - [→ *Topic 7-8*] Focus of stabilization policy

Natural rate = Frictional + Structural (excludes cyclical)

Summary: Unemployment

Key concepts:

1. Unemployment rate = Unemployed / Labor force
2. To be unemployed: not working + available + actively seeking
3. Natural rate ($\bar{u}$): Long-run unemployment (frictional + structural)
4. Three types: Frictional, Structural, Cyclical
5. Spain has structurally higher unemployment than most EU countries

[→ *Topic 7-8*] Cyclical unemployment and the Phillips curve are central to stabilization policy

Topic 1: Key Takeaways

The three main macroeconomic indicators:

1. **GDP** – Measures total production
   - $Y = C + I + G + NX$
   - Real GDP removes inflation
   - Useful but incomplete measure of well-being
2. **Inflation** – Measures rising prices
   - CPI: Fixed basket of consumer goods
   - Use to adjust values across time
   - Real interest rate = Nominal rate - Inflation
3. **Unemployment** – Measures unused labor
   - Natural rate = frictional + structural
   - Spain has structurally high unemployment

Next topic: What determines long-run GDP growth?

Thank you!

Questions?

Next: Topic 3 – Production & Growth